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Multi-view change point detection in dynamic networks

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ABSTRACT

Change point detection aims to find the locations of sudden changes in the network structure, which persist with time. However, most current methods usually focus on how to accurately detect change points, without providing deeper insight into the cause of the change. In this study, we propose the multi-view feature interpretable change point detection method (MICPD), which is based on a vector autoregressive (VAR) model to encode high-dimensional network data into a low-dimensional representation, and locate change points by tracking the evolution of multiple targets and their interactions across the whole timeline. According to the evolutionary nature of dynamic networks, we define a categorization of different types of changes which can occur in dynamic networks. We compare the performance of our method with state-of-the-art methods on four synthetic datasets and the world trade dataset. Experimental results show that our method achieves well in most cases.

1. Introduction

Change point detection is a very popular branch in the field of data mining and pattern recognition that discovers rare occurrences in rich data, because what stands out in the data is more major and interesting than understanding its general structure. Its task is to mark a time point at which the network structure has changed remarkably compared with the previous time snapshot, and the difference remains until the next change point [17]. There are many useful and practical application of change point detection across multiple domains. A small sample includes: intrusion detection for System [13]; identifying abnormal objects in social networks [35]; opinion deception and reviews spam [27] and pathology detection in medical imaging [2].

Soundarajan et al. [33] define dynamic networks as continuous time snapshots, each representing the entities (as vertices) and their connections (as edges) at a given moment. Examples of dynamic networks, include social networks [24,26], trade networks [8], transcriptional regulation networks [32], Scientific collaboration networks [25]. Due to the complexity of dynamic networks, there are various reasons for changes, such as abnormal evolution behaviors of nodes, communities, global level. For example, Somen Chakrabarti changes his research field from "parallel level systems" to "data and information systems" during the period from 1997 to 1998 [16]. The interpretability analysis of change point detection helps to better comprehend network evolution, thereby formulating effective interventions to cope with possibility risks in the networks. It can be undertaken to practices such as traffic prediction, intrusion detection and social network analysis.

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Existing change point detection methods are divided into two categories, namely, generative methods and feature-based methods, which are unanimously recognized by most researchers. They all focus on how to accurately detect the change points, without explaining the mechanism of its occurrence. For example, the Generalized Hierarchical Random Graph Model (GHRG) [29] delineates the change point detection problem as capturing the times when the parameter values of the generative model fluctuate significantly. CICPD [42] detects the change points through extracting node-level features. In network evolution, it is more interesting to explain the change reasons, which is conducive to in-depth understanding of internal incentives, so as to accurately formulate different countermeasures. In addition to the collective response to external events, the change points may also be related to macro, mesoscopic and micro changes in the network. Therefore, considering a single perspective is not enough to assist the perfect mining of the regular pattern of network evolution, and the networks should be represented comprehensively. We seek an efficient solution to not only locate change points but also explain why it happened.

Mining change points in dynamic networks caused by a single cause is not enough to fully explore the rules of networks evolution, that is, changes from different views are closely related. However, these methods [14,34,17,18,41] only consider some but not all changes in views, and which cannot model change points causes to achieve enhanced detection understanding. We propose a new multi-level feature interpretable change point detection model (MICPD), which improves the detection accuracy by mining the multi-view changes of the network and their interrelationships, and reveals the reason for the change point detection. Specifically, we first propose general categories of network changes and give examples of how each situation might occur. Then, we extract representation features for each category change separately, which track the historical evolution behavior of the network from node, community and global perspectives. Finally, we judge whether a change point occurred or not based on the predicted error rate. Our main contributions are three-fold:

- We introduce a multi-view feature interpretable change point detection method (MICPD), which encodes high-dimensional network data into a low-dimensional representation based on a vector autoregressive (VAR) model, and locates change points by tracking the evolution of multiple objects and their interactions across the whole timeline.
- MICPD explicitly captures node, community, and global changes to simulate microscopic, mesoscopic, and macroscopic changes in dynamic networks. It is a new change point detection framework and has good interpretability.
- We evaluate the performance of the MICPD method on four synthetic datasets and one real dataset. Compared with advanced methods, all experimental results demonstrate the capability and consistency of our method in dynamic network change point detection.

The rest of this article is as follows. Section 2 describes related work on change point detection. Section 3 formalizes the change point detection in dynamic network. Section 4 details our proposed method. In Section 5, we use five datasets to validate and analyze our method. Finally, we give concluding remarks and a discussion of future work in the Section 6.

2. Related work

2.1. Generative methods

The idea of the generative methods is to fit the dynamic network with a specific probabilistic model, so as to capture change points by using the model parameter floating. These generative methods differ in using different probabilistic model types.

Peel and Clauset [29] presents GHRG, which uses generalized hierarchical random graph models and Bayesian hypothesis testing to quantitatively detect when change points occur. Cheung et al. [8] fits the dynamic network through the Stochastic Block Model, and an objective function based on Minimum Description Length Principle realizes the simultaneous change point detection and community structure. Wang et al. [36] introduce a conditionally independent two-state Markov chain to represent the state transitions of edges in adjacent time slices. It can detect changes in parameter values of a given model and changes in model types. Wang et al. [37] propose to capture changes between communities and within communities by compressing the network. Here, each super node is a community, so that the time interaction of the community can be simulated through a super network. Li et al. [23] combine change point detection and community evolution. The former uses a hierarchical random graph to capture the mutation of the network structure, and the latter uses community evolution events to explain the common features caused by the mutation. Jiao et al. [19] assume that the generation process of the dynamic network obeys the stochastic block model (SBM), and calculate the change point through the node-level popularity and the transition parameters of the community.

Using a particular probabilistic model is often problematic. If the data does not match, they are probable to achieve lower performance. In addition, dynamic networks structure is complex and running time is limited, so most probabilistic models can only be applied to specific data structures.

2.2. Feature-based methods

Feature-based methods extract network features per time slice and compute the gap between them. Its main components are feature extraction and change detection strategies.

Zhu et al. [42] uses the nodes' importance to characterize networks, and calculated the distance between two networks to detect change points. Kendrick et al. [20] employ different global network metrics (Size, Density, Average Clustering Coefficient, Average Shortest Path) to achieve change point detection. Darst et al. [10] aim to create time slices to capture the time-scale evolution characteristics of dynamical systems. The method iteratively builds time intervals based on the similarity measure of the current set of events and the next event. The approach treats the presence or absence of a dynamic network edge as a special event that can be tailored to discover change points in dynamic networks. Chen et al. [7] build a coding and decoding framework. It uses matrix factorization technique to unsupervisedly divide the dynamic process of the network into different sub-sequences. Each sub-sequence is composed of a network, thus making the dynamic networks more concise. Huang et al. [18] present an anomaly detection algorithm founded on graph Laplacian. It simulates the long-term and short-term dependencies of dynamic evolution using two sliding windows to capture both gradual and abrupt changes. De Ryck, De Vos and Bertrand [11] notice the non-independent identically distributed nature of the data, using discrete Fourier transforms to transform time-domain features into frequency-domain space, and combining them to capture time-invariant features. Anomaly Transformer [39] assumes that the distance between the prior-association and the series-association of abnormal data is small, while the normal data is the opposite. It adopts Gaussian kernel and Transformer to model the prior-association and series-association, respectively, to reflect the association discrepancy.

Compared to the generative methods, the feature-based method has lower complexity because there is no parameter estimation step. How to choose a suitable set of feature vectors for change point detection is a notable issue.

The above methods only pay attention to detect change points, and do not explain the mechanism of its occurrence. Therefore, we propose a unified framework to simulate different views of change points and capture the mechanisms they produce.

3. Problem definition

In order to detect change points efficiently, accurately and interpretably, it is required to reasonably represent dynamic networks and define the problem. Symbols and formal definitions will be introduced below.

We symbolize the dynamic networks as a series of snapshots, i.e. $G = \{G^1, G^2, \cdots, G^T\}$, where T is the amount of snapshots. Each time snapshot G^t consists of a vertex set V^t and an edge set E^t. It can be specifically expressed as an $|V| * |V|$ dimensional binary adjacency matrix A. The relationship between the two vertices i, j is expressed as the corresponding matrix element $a_{ij} = 1$, otherwise $a_{ij} = 0$. There is no constraint on the amount of nodes in each time slice, which means that time snapshots G^t and $G^{t'}$ can have different nodes, where t is not equal to t'. This assumption is very consistent with the evolutionary rules of the network, because the nodes that are active at this moment may also disappear at the next moment. For example, in world trade dataset [15], each time slice G represents the trade volume of two countries within one year, the node represents the country, and the connection between the nodes indicates that there is trade exchange between the two countries, and there is at least a communication. We use $C^t = \{c_1^t, c_2^t, \cdots, c_k^t\}$ to represent the community membership and k represents the number of communities detected in time slice t.

The change point detection in dynamic networks is usually to seek the time t, which triggers significant structural changes of the networks at different view. The evolutionary patterns of dynamic networks are invisible, so we need to devise a way to formalize the change points. It is worth noting that the evolutionary behavior changes from different perspectives of the networks are the fundamental reason for changes emergence. Based on above, we focus on detecting the change points caused by the structural changes of different views in the network.

Problem Definition: Given a network sequence $G = \{G^1, G^2, \cdots, G^T\}$, we define a change scoring function $f = \alpha f_n + \beta f_c + \gamma f_g$, where f_n, f_c, f_g are used to calculate nodes, communities, and global change scores, respectively. Our goal is to find a set $S \subset \{1, 2, \cdots, T\}$, such that $t \in S \Leftrightarrow f^t \geq th_1$ and $f^{t+1} \leq th_2$, for some pre-specified threshold th_1, th_2.

Based on this, we propose a multi-view interpretable change point detection method. Over successive time intervals w, it predicts a multi-view compact representation of temporal snapshots with strong statistical correlations. Dynamic network change points can be identified when prediction errors increase substantially. Our proposed method first slices the dynamic networks and adopts the GED [5] algorithm to capture the community evolution behavior. Here, we use the SSRM [1] algorithm to identify mediators, which are bridges between two communities and correspond to global changes in the network. Second, we compress high-dimensional data into different representation spaces to extract node, network, and global features. Finally, the Vector Autoregression (VAR) model [22] is used to predict the future development of time series networks. We hypothesize that the multi-view features in the non-smooth stage of network evolution are significantly different, which leads to a transient increase in prediction error. As the prediction model adapts to the new stage, the prediction error gradually decreases. Therefore, the probability of a network change point occurring increases as the prediction error increases. Fig. 1 shows the overall architecture of the MICPD. In the next section, it is explained in detail.

4. Method framework

In this section, we first consider the network evolution forms from different perspectives (e.g., nodes, communities, globals) to ensure that most possible changes are captured. Based on this, we propose a general categorization of change-types in dynamic networks. Then, we use the vector autoregressive model to calculate the node, community, and global change scores respectively. Finally, we integrate multi-view changes into a framework to enable change point detection.

4.1. Multi-view changes categorization

In previous studies, the evolution of dynamic networks was monitored through a single perspective, but the non-smoothness of its evolution was manifested in the changes of different perspectives. Although some methods define the type of change point, the specific implementation does not distinguish the category. This also leads to the occurrence of change points becoming inexplicable.

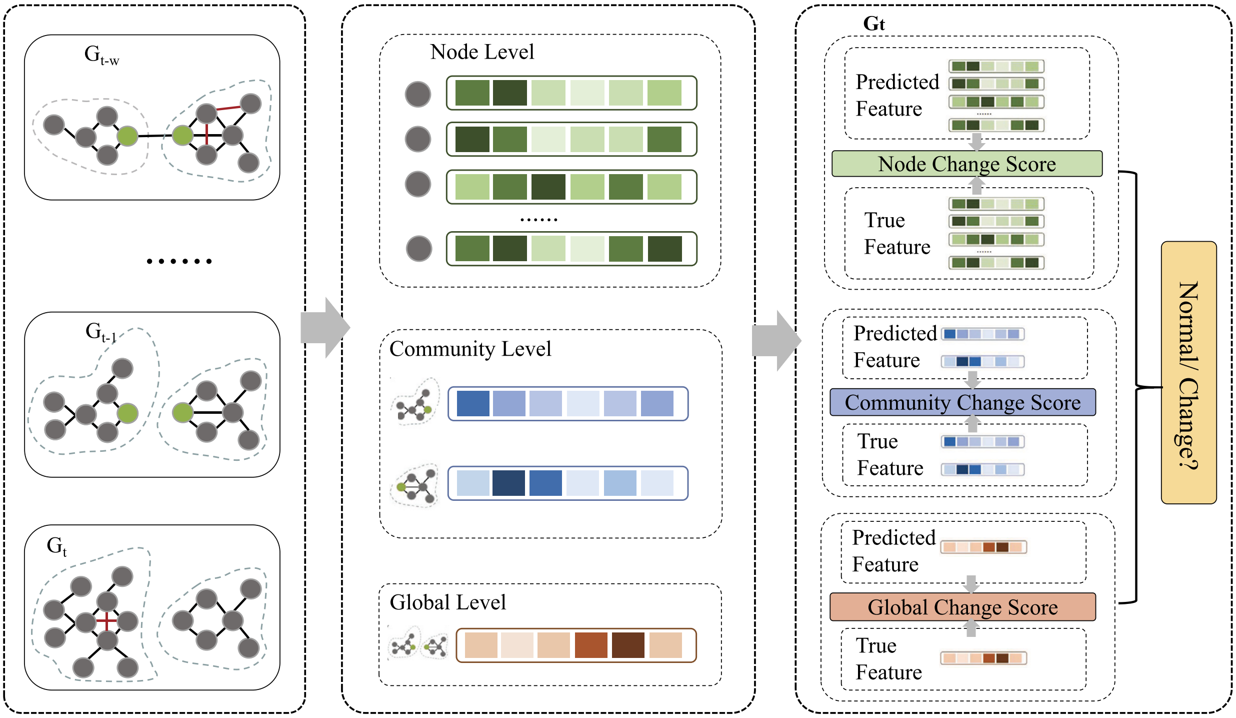


Fig. 1. Outline frame. First, the dynamic network is divided into consecutive temporal snapshots, and community discovery and mediators identification are performed on each snapshot. Next, we extract features from three perspectives: nodes, community, and global. Finally, the VAR model is used to predict the multi-view features at time t by tracking node, community, global evolution behavior from time $t-1$ to time $t-w$. We compute the difference between predicted and true values and use hypothesis testing to enable interpretable change point detection.

We learn a more informative representation of the network from multiple views, which contributes to the interpretability of detection and the improvement of accuracy. It is also a challenging task to employ more information to discover the statistical analysis concepts. On this basis, we classify and define changes in dynamic networks from multiple views, which are completely different from the definitions in previous studies. Each case considers a specific network perspective and describes the changes that may be caused by it.

A. Node Changes

At present, most of the research focuses on extracting macroscopic features to achieve change point detection, but fail to analyze the nodes with structural changes at the microscopic level. In this research, we define node changes as significant changes compared to usual or past states in dynamic networks. The structure characteristics of the node can accurately capture the abnormal evolution behavior of the node. An example is shown in Fig. 2, where the structure of parts of the network is more dense or sparse due to the increase or decrease of some nodes connecting edges.

Examples for these situations are terrorist communication networks [21], where increased or decreased communication between participants could mean a terrorist attack in the future, or a sudden increase in links between cities in a transportation networks [3], meaning holidays are coming.

B. Community Changes

Each group that exists in a complex network is called a community, that is, the internal links are close, and the external links are sparse. By analyzing the network time series, Dakiche et al. [9] observe that communities are constantly evolving (continuing, forming, dying, merging, splitting). A large number of communities have evolved relatively smoothly, and their average value defines the evolutionary trend of dynamic networks. However, some objects evolve very distinct behaviors than others, and we collectively refer to them as anomalous communities. An example is shown in Fig. 2. One of the original communities grew, and the other disappeared. At the same time, a new community appeared, causing major changes within the network.

Examples for these situations are bibliographic networks, one conference changes the direction of a subtopic, while other conferences in the field remain constant [16].

C. Global Changes

Global changes are defined as changes in the links between communities. Mediators are individuals who act an significant contribution in linking communities throughout the networks [1]. Compared with the non-change points network evolution, the overall number of mediators and their links increase or decrease significantly and thus the network changes. Fig. 2 shows a minimal example for ease of explanation. The connection established between C^1 and C^2 causes the entire network to change from a disconnected graph to a connected graph.

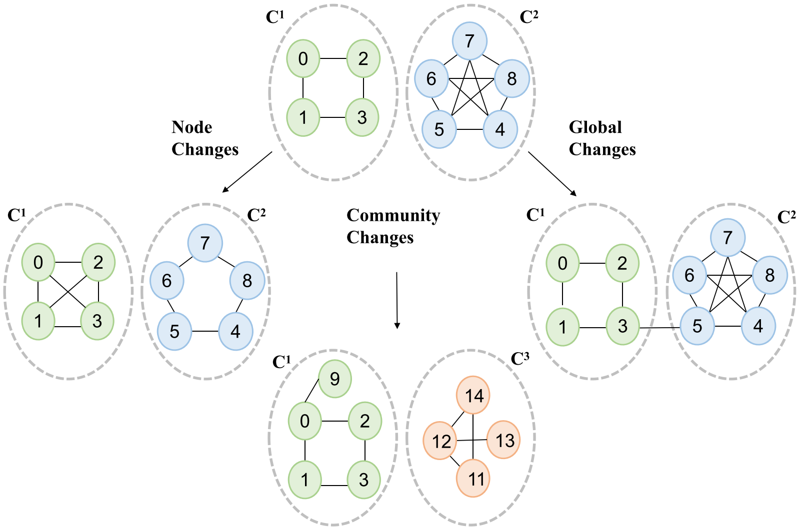


Fig. 2. Example to demonstrate multi-perspective changes on a two-community network. The increase or decrease of some node links is treated as a node structure change. The C^1 community grows, the C^2 community disappears, and the C^3 community forms, which mainly affects the community structure, so it corresponds to community changes. The relationship between C^1 and C^2 gives rise to the connectedness of the whole network, corresponding to the global changes.

Examples for these situations are email network, where multiple groups work together on a funding proposal that will cause an email explosion among communities, or a sudden increase or decrease in communication between communities in a social network could indicate that something is about to happen.

4.2. Multi-view change detection

To clarify the reasons for the change points, we explore the historical evolution behavior of the networks from different perspectives, that is, structural changes across time lines. Node-centric changes capture the historical behavior of each node in the networks. Both community-centric Changes and global-centric changes take community information into account. The former focuses on the dynamic evolution of each community, while the latter captures dynamic interactions between communities. Based on the above, multi-view changes are integrated into a framework to implement change point detection. In what follows, we detail the steps.

A. Node-Centric Changes

Node-centric changes extraction includes two main steps: (1) Extract the structural features of each node in the dynamic network. (2) Calculate the node change scores by tracking the historical behavior of nodes.

We use a representative node feature matrix $F_{i^t} = \{f^1_{i^t}, f^2_{i^t}, \cdots, f^m_{i^t}\}$ to characterize the observed behavior of node i, where m is the number of picked features, t represents time. Accordingly, each graph snapshot G^t is represented as a feature matrix $F_{N^t} = \begin{pmatrix} f^1_{1^t}, f^2_{1^t}, \cdots, f^m_{1^t} \\ f^1_{2^t}, f^2_{3^t}, \cdots, f^m_{3^t} \\ \cdots \\ f^1_{n^t}, f^2_{n^t}, \cdots, f^m_{n^t} \end{pmatrix}$, including n number of nodes. According to previous work [31], we represent each node using six node features: degree, betweenness, closeness, clustering coefficient, eigenvector and PageRank.

History-based node changes: we employ Vector Autoregression (VAR) model [22] to track the evolutionary behavior of each node in the dynamic networks. This method treats each feature variable in the time series data as the lag value of all variables. Thus, the univariate autoregressive model is generalized to the multivariate vector autoregressive model. More importantly, the model generally yields superior predictive performance on time series data and is better to train. Since the $i-th$ row of the feature matrix F_{N^t} stores the information of the i node, its historical behavior can be described as follows:

$$F_{i^t} = c + B_1 F_{i^{t-1}} + B_2 F_{i^{t-2}} + \ldots + B_j F_{i^{t-j}} + \ldots + B_w F_{i^{t-w}} + e^t, \tag{1}$$

where B_j is a $m * m$ matrix representing the node feature conversions between time t and $t-j$, and c is a $m * 1$ vector of constants, e_t is $m * 1$ error vector. w is the lag order.

Finally, the change score for a given node is defined as the difference between the true and predicted eigenvectors. Eq. (1) predicts the feature vector of the $i-th$ node at time t as F_{i^t}. Jensen-Shannon (JS) divergence is employed to calculate the distance between the true feature $\hat{F}_{i^t}$ and the predicted feature F_{i^t} as the change score for node i, as shown below:

$$JS(\hat{F}_{i^t}||F_{i^t}) = \frac{1}{2}\left\{ D_{KL}(\hat{F}_{i^t}||\frac{F_{i^t} + \hat{F}_{i^t}}{2}) + D_{KL}(F_{i^t}||\frac{F_{i^t} + \hat{F}_{i^t}}{2}) \right\}. \tag{2}$$

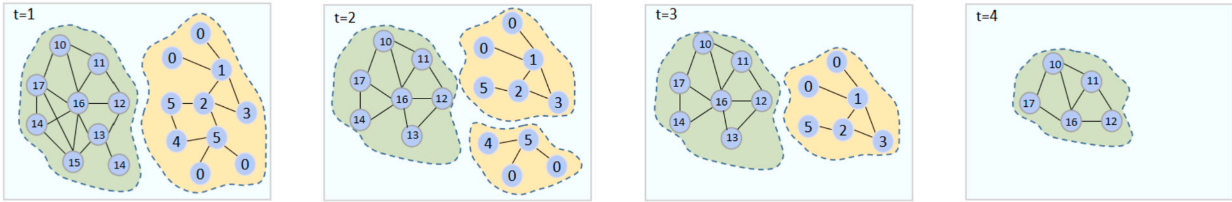


Fig. 3. Communities evolution.

Here, the change score of node i at time t is defined as $JS(\hat{F}_{i^t}||F_{i^t})$. The total nodes change score is $f_n^t = \sum_{i=1}^{n} JS(\hat{F}_{i^t}||F_{i^t})$ in t timestamp.

B. Community-Centric Changes

Capturing community-centered changes involve three main steps: (1) Split the dynamic network into snapshots, and perform static community detection on each snapshot. (2) The GED method [5] is used to discover community evolution. (3) The vector autoregression (VAR) model [22] locates community changes based on historical behavior.

The community detection algorithm divides the time snapshot G^t into blocks, so that the internal links are relatively dense, and the external links are relatively sparse. At present, there are many clustering algorithms that can be used to discover communities. The Louvain algorithm [4] is a very efficient non-overlapping community detection algorithm that runs in essentially linear time on some real-world networks. Owing to the large scale of our databases, this algorithm is the most proper for our requirements. We use the set $C^t = \{c_1^t, c_2^t, \ldots, c_k^t\}$ to denote that k communities are detected in time t. Each community $c_k^t \in C^t$ represented by a graph $G_k^t = (V_k^t, E_k^t)$, which is a subgraph of G^t with vertex set $V_k^t \in V^t$, and edge set $E_k^t = \{(u,v) \in E_t : u, v \in V_k^t\}$.

The community tracker provides community evolution events in consecutive time snapshots. We find a set of communities $C^t = \{c_1^t, c_2^t, \cdots, c_k^t\}$ in snapshot G^t and their ancestor set $C^{t-1} = \{c_1^{t-1}, c_2^{t-1}, \cdots, c_k^{t-1}\}$ in graph snapshot G^{t-1}. In this paper, the GED [5] method based on similarity measure accurately discovers community evolution. This method can detect seven types of community evolution events: formation, survival, merger, division, dissolution, growth, and contraction. It is formulated using two quantitative and qualitative factors as follows

$$I(G^{t-1}, G^t) = \frac{|G^{t-1} \cap G^t|}{|G^{t-1}|} \frac{\sum_{x \in (G^{t-1} \cap G^t)} SP_{G^{t-1}}(x)}{\sum_{x \in (G^{t-1})} SP_{G^{t-1}}(x)}. \tag{3}$$

According to this definition, communities will be paired if the $I(G^{t-1}, G^t)$ value exceeds the threshold $\theta \in [0,1]$, which means that they have a higher proportion of common nodes.

The communities evolutionary is highly dynamic. For a community c_a^t, we consider not only the community set c_a^{t-1} found at the previous timestamp $t-1$, but also all community sets $\{c_a^{t-1}, c_a^{t-2}, \cdots, c_a^{t-w}\}$ detected in the last w step. After the communities on different timestamps are matched together, the evolution information of the community is obtained, as shown in Fig. 3. We find all connected parts and define them as community evolution, which are displayed with different colors.

Based on the above, a representative set of features is used to describe the observed behavior of a given community. We represent the community c_a^t in time slice G^t as a feature matrix $F_{c_a^t} = \{f_{c_a^t}^1, f_{c_a^t}^2, \cdots, f_{c_a^t}^m\}$, where $f_{c_a^t}^1$ denotes first eigenvalue of c_a^t at time t. m is the number of features chosen. According to previous work [28], six characteristics are used to describe communities features: intra-community edge, inter-community edge, average clustering coefficient, density, aging, and activity.

History-based community changes: Likewise, we use the VAR to track the evolutionary behavior of each community in the dynamic networks. We describe the community c_a^t features as follows:

$$F_{c_a^t} = c + A_1 F_{c_a^{t-1}} + A_2 F_{c_a^{t-2}} + \ldots + A_j F_{c_a^{t-j}} + \ldots + A_w F_{c_a^{t-w}} + e^t, \tag{4}$$

where A_j is a $m * m$ matrix describing the community conversions between timestamp t and $t-j$. The VAR model is a dynamic model for analyzing the variables in the joint without any constraints.

In the last step, we also use Jensen-Shannon(JS) divergence to calculate the distinction between the true feature eigenvectors and predicted feature eigenvectors of the c_a^t community, called the community c_a^t change score. From the above, the total communitys change score at time t as $f_c^t = \sum_{a=1}^{k} JS(\hat{F}_{c_a^t}||F_{c_a^t})$.

C. Global-Centric Changes

Capturing global changes mainly includes two main steps: (1) Identify the factors that cause global changes and extract corresponding features. (2) Calculate the global change score.

According to section 4, mediators play the role of connecting different communities in the network. We use SSRM [1] to identify the mediators that establish links for the community in the network and track them. Here, we count the number of shortest paths through nodes between different communities, the C-Betweenness Centrality (CBC) indicator, to identify mediators. CBC for node v is:

$$CBC(v) = \frac{1}{2} \sum_{p \in Cpaths} I_p(p, v). \tag{5}$$

Here, $CPaths = \{p | c_{s_p} \neq c_{e_p}\}$ represents the shortest path set of the network, which connects two different communities, where s_p and e_p indicate the start and end respectively. The community attribution of the v node is c_v. Set $I_p(p, v)$ to 1 if node v is on path p, otherwise 0. According to the special properties of complex networks, the mediation distribution is close to a power-law distribution. Therefore, nodes at the tail of the mediation distribution mean connecting more different communities, and these nodes are mediators.

We represent the global feature on the time slice G^t as a feature matrix $F_{g^t} = \{f^1_{g^t}, f^2_{g^t}, \cdots, f^m_{g^t}\}$, where $f^i_{g^t}$ is the $i - th$ eigenvalue of the global network attribute. We adopt three global features: mediators number, mediators average degree, mediators average shortest path.

History-based global changes: We also use Vector Autoregression (VAR) to model the global historical behavior of the network. The feature matrix F_{g^t} to represent the global attributes of the network at time t, and describe its evolution as follows:

$$F_{g^t} = c + Z_1 F_{g^{t-1}} + Z_2 F_{g^{t-2}} + \ldots + Z_j F_{g^{t-j}} + \ldots + Z_w F_{g^{t-w}} + e^t, \tag{6}$$

where Z_j is a $m * m$ matrix describing the network global conversions between time t and $t - j$. Similarly, we define the outlier of the global network at timestamp t as the difference between the true feature eigenvectors and the predicted feature eigenvectors. The total global change score is $f^t_g = JS(\hat{F}_{g^t} || F_{g^t})$ in t timestamp.

4.3. Integrated framework

Above, the multi-perspective changes of dynamic networks are defined: node changes, community changes, and global changes. These three changes combine into a comprehensive framework:

$$f^t = \alpha f^t_n + \beta f^t_c + \gamma f^t_g \qquad s.t. \qquad \begin{cases} f^t > th_1 \\ f^{t+1} < th_2 \quad \text{(time t is a change point)}. \end{cases} \tag{7}$$

Here, we have N time snapshots and get $N - 1$ change scores. How to pick two appropriate thresholds to judge in which time slice the network changes? This paper employs a permutation test based method to judge the threshold. For a desired significance level α, use $100\alpha\%$ quantiles as the threshold.

5. Experiments

In this section, we conduct sufficient experiments on three different types of datasets to verify the effectiveness of the MICPD framework. We contrast it with state-of-the-art baseline models on the change point detection task and perform parameter sensitivity analysis on MICPD. All codes are available in GitHub.[1]

5.1. Baseline methods

We choose 4 change point detection methods as baselines to verify the effectiveness of MICPD. They are listed as follows:

- **CICPD [42]**: This method extracts graph features for each temporal snapshot, and thus construct a feature vector per temporal snapshot. Spectral clustering calculates the similarity between feature vectors to achieve change point detection.
- **LAD [18]**: This method calculates the singular value decomposition of the Laplacian matrix for each time snapshot, and selects the first K singular values to identify the change point.
- $TS - CP^2$ **[12]**: This method exploits contrastive learning to detect the change points, and adopts temporal convolutional networks (TCN) to learn the compact representation of time series.
- **NetWalk [40]**: It detects change points by learning dynamic network representations. It proposes clique embeddings, which collectively minimize the pairwise distances between vertex representations in each random walk.

5.2. Synthetic experiments

In order to verify the performance of MICPD, we generate four synthetic datasets, which are node change dataset (NCD), community change dataset (CCD), global change dataset (GCD), and hybrid change dataset (HCD). Currently, many methods can be used to generate dynamic networks with community structures, such as RDyn [30] and DSABM [6]. We choose a typical Stochastic Block Model (SBM) [38] for data generation. Each dataset is a sequential network with 500 nodes and 100 snapshots. Different snapshot pairs of the same network are not independent, each snapshot relies on the previous. Here N_c represents the number of communities in the snapshot, and p_{ex}, p_{in} indicate the external and internal community connectivity probability respectively. Global change is an obvious behavior, simulated by associations between communities, the magnitude of which indicates the intensity of global change. A larger external community connectivity probability p_{ex} implies stronger connectivity between communities, and vice versa. For example, at the $16th$ time snapshot in the global changes dataset, we set the external community connectivity

[1] https://github.com/xieyingjie/MICPD.

Table 1
The generation process of four synthetic datasets (node changes, community changes, global changes, hybrid changes).

Time	Hybrid changes (HC)		Node changes (NC)	Global changes (GC)	Community changes (CC)
	Type	Generative SBM Model	Generative SBM Model	Generative SBM Model	Generative SBM Model
0	start	$N_c = 4$, $p_{in} = 0.45$, $p_{ex} = 0.05$	$N_c = 4$, $p_{in} = 0.45$, $p_{ex} = 0.05$	$N_c = 4$, $p_{in} = 0.45$, $p_{ex} = 0.05$	$N_c = 4$, $p_{in} = 0.45$, $p_{ex} = 0.05$
16	NC & CC	$N_c = 8$, $p_{in} = 0.55$, $p_{ex} = 0.05$	$N_c = 4$, $p_{in} = 0.65$, $p_{ex} = 0.05$	$N_c = 4$, $p_{in} = 0.45$, $p_{ex} = 0.15$	$N_c = 10$, $p_{in} = 0.45$, $p_{ex} = 0.05$
34	NC & GC	$N_c = 8$, $p_{in} = 0.45$, $p_{ex} = 0.10$	$N_c = 4$, $p_{in} = 0.55$, $p_{ex} = 0.05$	$N_c = 4$, $p_{in} = 0.45$, $p_{ex} = 0.10$	$N_c = 8$, $p_{in} = 0.45$, $p_{ex} = 0.05$
61	GC & CC	$N_c = 10$, $p_{in} = 0.45$, $p_{ex} = 0.05$	$N_c = 4$, $p_{in} = 0.35$, $p_{ex} = 0.05$	$N_c = 4$, $p_{in} = 0.45$, $p_{ex} = 0.05$	$N_c = 6$, $p_{in} = 0.45$, $p_{ex} = 0.05$
87	ALL	$N_c = 4$, $p_{in} = 0.55$, $p_{ex} = 0.10$	$N_c = 4$, $p_{in} = 0.45$, $p_{ex} = 0.05$	$N_c = 4$, $p_{in} = 0.45$, $p_{ex} = 0.25$	$N_c = 4$, $p_{in} = 0.45$, $p_{ex} = 0.05$

Table 2
Precision, Recall and F1 of MICPD and the baseline methods.

Time	Hybrid changes (HC)			Node changes (NC)			Global changes (GC)			Community changes (CC)		
	Precision	Recall	F_1	Precision	Recall	F_1	Precision	Recall	F_1	Precision	Recall	F_1
CICPD	0.6666	0.5000	0.5714	0.6666	0.5000	0.5714	0.5000	0.2500	0.3333	0.6666	0.5000	0.5714
LAD	0.8000	**1.0000**	**0.8888**	0.6000	0.7500	0.6666	0.8000	**1.0000**	**0.8888**	0.6000	0.7500	0.6666
NetWalk	**1.0000**	0.7500	0.8571	**0.7500**	0.7500	0.7500	0.5000	0.2500	0.3333	**0.8000**	**1.0000**	**0.8888**
TS-CP^2	0.6000	0.7500	0.6666	0.6666	**1.0000**	0.8000	**1.0000**	0.5000	0.6666	0.6666	0.5000	0.5714
AJ	0.2000	0.2500	0.2222	0.3333	0.5000	0.4000	**1.0000**	0.2500	0.4000	0.5000	0.5000	0.5000
MICPD	0.8000	**1.0000**	**0.8888**	0.6666	**1.0000**	**0.8000**	0.6666	0.5000	0.5714	0.6666	**1.0000**	0.8000

probability p_{ex} change from 0.05 to 0.15. Community changes correspond to the occurrence of community evolution events. This change is simulated by changing the number of communities, which can correspond to the sudden merging of two communities into one community. The larger the number of communities N_c, the more communities the current snapshot has. At time slice 34 of the community change dataset, we set the N_c from 10 to 8. Node changes are made by changing the value of the internal community connectivity probability p_{in}. The larger its value is, the stronger the node connectivity is. At time slice 61 of the node change dataset, we shift the internal community connectivity probability p_{in} from 0.55 to 0.35. It is worth noting that the hybrid change dataset is generated by combining the three changes with each other. The other three datasets each correspond to only one change. The details of the generation process are shown in Table 1.

We evaluate the effectiveness of the MICPD method on four synthetic datasets by comparing the detected change points with the ground-truth of the outliers we inject. In addition to the adopted four baselines, we also add a true naive baseline that only utilizes cosine similarity to compute distances between snapshots. The main results are reported in Table 2. In most cases, it can be seen that our method MICPD significantly outperforms the baseline methods in measures F1 and recall, which implies that multi-view features can capture network structural properties more sensitively. Fig. 4 presents a closer inspection of the change points detected by each method on the node change dataset. CICPD only picked out 2 change points corresponding to heavy node changes, while LAD, NetWalk missed change point 2, which belongs to a slight change. In contrast, both TS-CP^2 and our method MICPD detect all change points. For the community change dataset, as shown in Fig. 5. TS-CP^2 and CICPD detect only 2 change points and miss half of them. LAD failed to detect a slight community change point. Compared with the above methods, our method hits all change points and achieves better recall. But due to false positives, our method does not perform as well as NetWalk in terms of precision and F_1. The detailed results of the above methods on the global variation dataset are shown in Fig. 6. CICPD and NetWalk only detect 1 heavy node change and TS-CP^2 also performs poorly on recall. Although LAD has higher accuracy, our method also detects most of the change points. As to hybrid change dataset, where all three types of changes exist. From Fig. 7, we observe that MICPD is able to recover all change points perfectly. Obviously, the error rates of the baselines are higher than our proposed method, indicating that MICDP comprehensively captures changes in node, community and global structure, and it does not fully depend on any single change.

5.3. Real-world experiments

World trade dataset [15]: The dataset records the trade flows of 196 countries from 1948 to 2000, which includes the total annual export and import of both countries. An edge is formed between two countries if there is trade between them. We slice a annual dynamic social network from, which contains 196 nodes and 53 time slices. The ground truth events in this dataset have correspond to significant historical events affecting trade, such as Tariff change, establishment of economic community, Oil shock, as shown in Fig. 8.

Our proposed algorithm detects 6 change points on this dataset, as shown in Fig. 8. It can be found that the change point hit rate of MICPD is significantly better than the baselines, implying that it more comprehensively captures network structure changes. The four baseline methods either missed majority the events or identified many extra change points. Specifically, LAD, Netwalk, CICPD and $TS-CP^2$ only detected a part change points such as Africa started to get involved in trading, the collapse of the Soviet Union, the oil crisis and tariff reduction. MICPD identifies change points for many types of changes in the network, and detected events are highly consistent with predetermined events. According to Fig. 9, such as in 1960, the large parts of Africa, after being colonized to independence, got involved in trade, which led to the formation of many new communities. In 1967, our method detects that

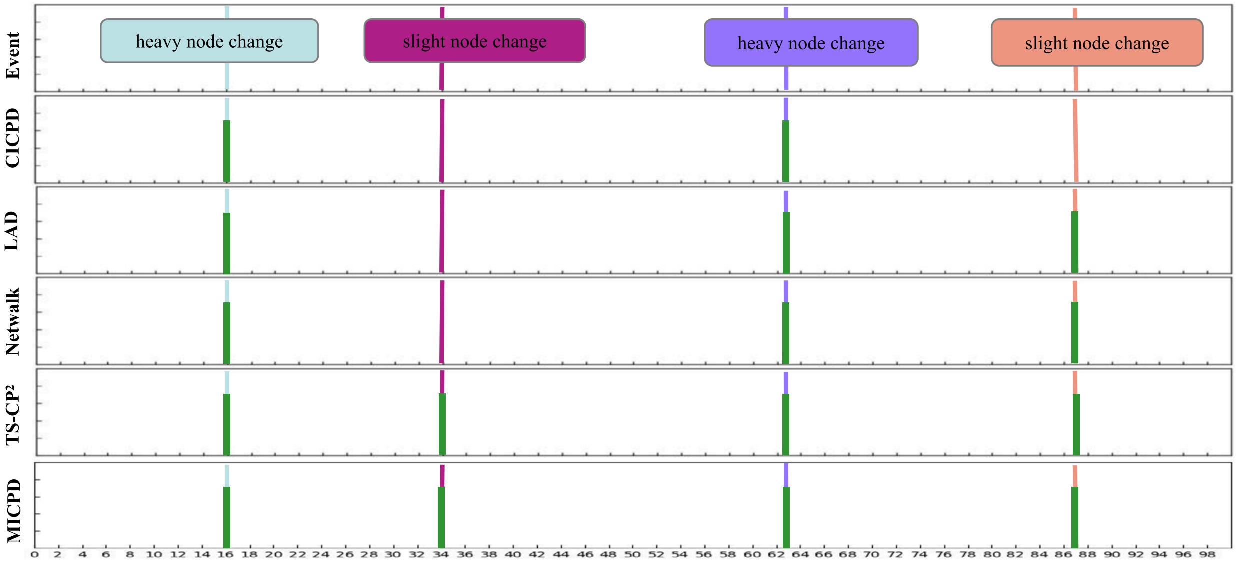


Fig. 4. Change point detection on the node change dataset. Detected change points are marked with green vertical bars. The events are marked in colored boxes associated with the colored lines spanning all methods.

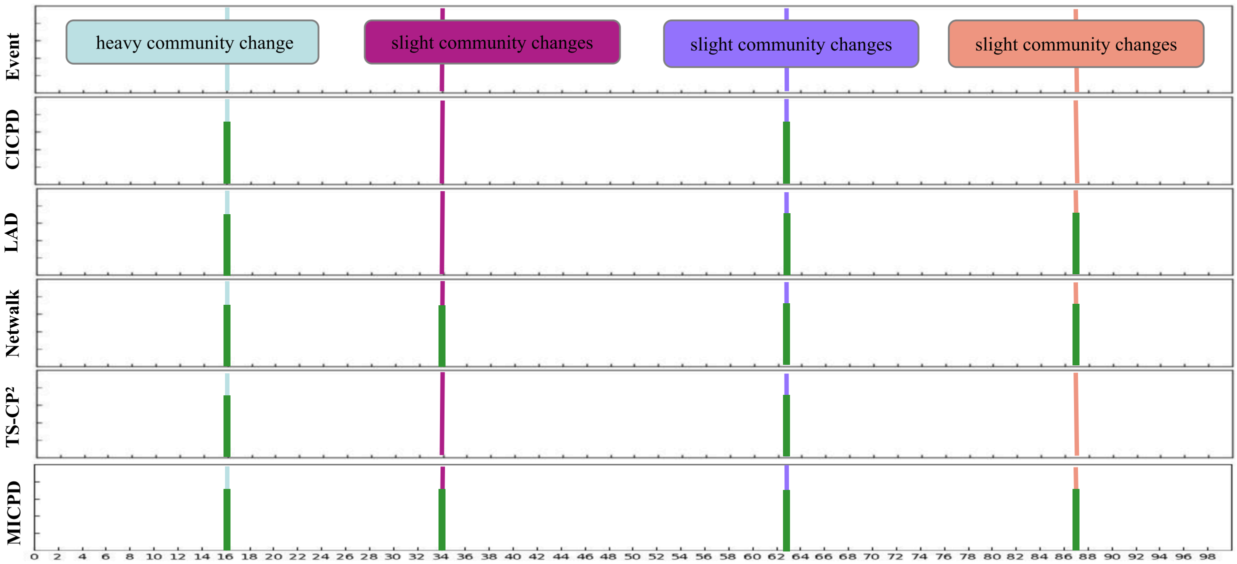


Fig. 5. Change point detection on the community change dataset. Detected change points are marked with green vertical bars. The events are marked in colored boxes associated with the colored lines spanning all methods.

the number of communities in the network changed from 4 in 1966 to 3, which corresponds to the establishment of the European Community in the real world. During the year, Barbados (BAR) traded with 28 countries, down from 35 in 1966. See Fig. 10 for details. The change score of communities and nodes rises sharply, so 1967 is detected as a change point. It can be seen that the occurrence of the change points in 1960 and 1967 was mainly due to the major changes in the community structure. At the node level, the 1973 oil crisis had a particularly severe impact on the major oil exporting countries, which saw a sharp decrease in the number of trade imports and exports. In 1983, with the recovery of diplomatic relations among countries and the gradual expansion of global economic integration ("globalization"), the volume of trade between several economies soared, resulting in stronger ties between communities than ever before. In conclusion, our framework not only accurately identifies change points in the world trade dataset, but also comprehensively explains what types of changes have caused them to occur.

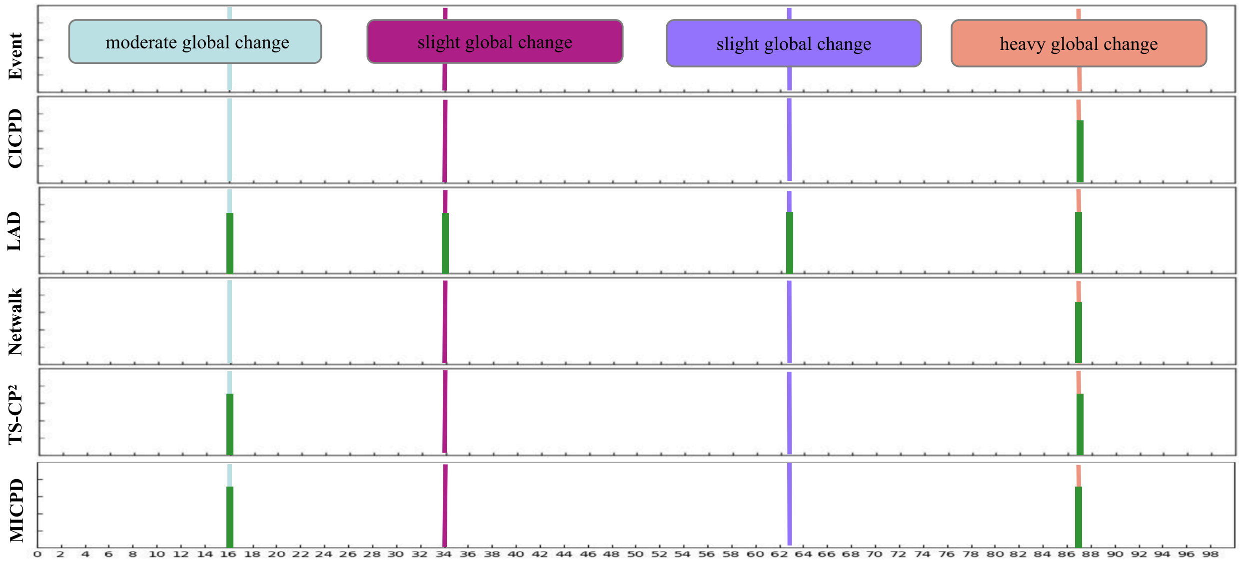


Fig. 6. Change point detection on the global change dataset. Detected change points are marked with green vertical bars. The events are marked in colored boxes associated with the colored lines spanning all methods.

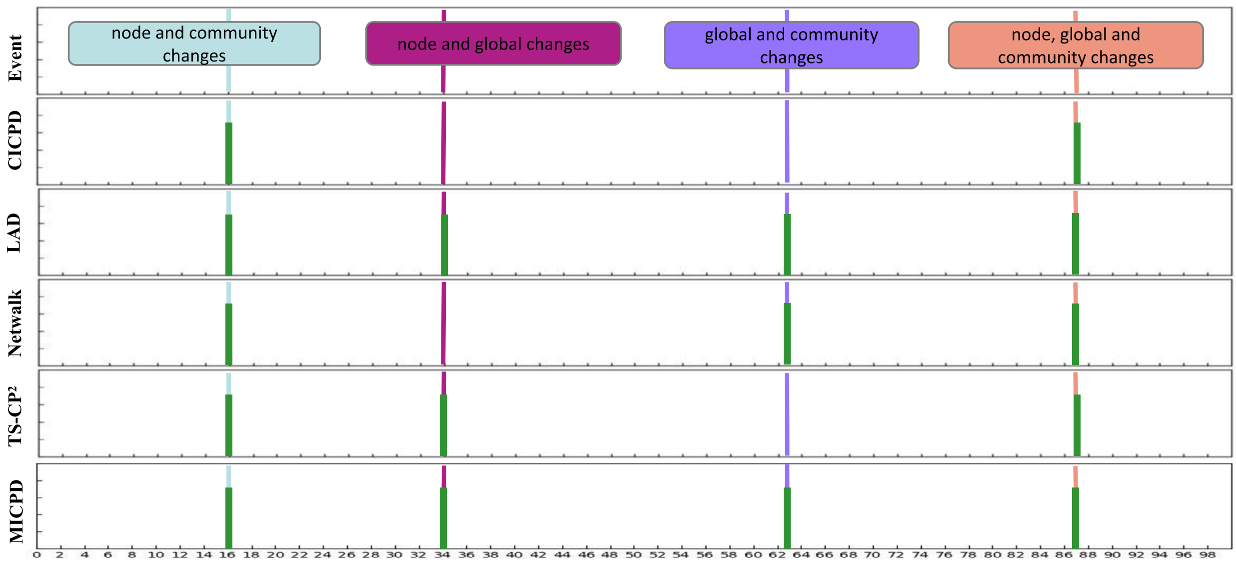


Fig. 7. Change point detection on the hybrid change dataset. Detected change points are marked with green vertical bars. The events are marked in colored boxes associated with the colored lines spanning all methods.

5.4. Parameter sensitivity

In order to verify how MICPD's hyperparameters affect performance, we vary the hyperparameters in the several experiments. There are two main hyperparameters in our proposed model: threshold th_1 and threshold th_2. We set hyperparameters $th_1 = 0.35$, $th_2 = 0.30$ as default for synthetic datasets and $th_1 = 0.40$, $th_2 = 0.30$ for World trade dataset. We select one parameter each time to change and fix the other parameters to check the impact of this parameter on the task of change point detection with Macro F_1 as a metric on five datasets.

It can be seen from Fig. 11(a) that as th_1 increases, the performance of the synthetic datasets becomes better, but when th_1 is greater than 0.35, it decreases instead. The world trade dataset also exhibits a similar situation, with the best performance when th_1 is around 0.40. The reason is that if th_1 is too small, there will be many false positives, and if it is too large, there will be many false negatives. From Fig. 11(b), we can observe that the performance of synthetic datasets gets better as th_2 gets larger. The world trade dataset has the best performance when th_2 is 0.30. If it is too small, many change points will be ignored. In the world trade dataset th_2 is too large, resulting in false positives (some snapshots change, but it doesn't persist), causing a slight drop in performance. The

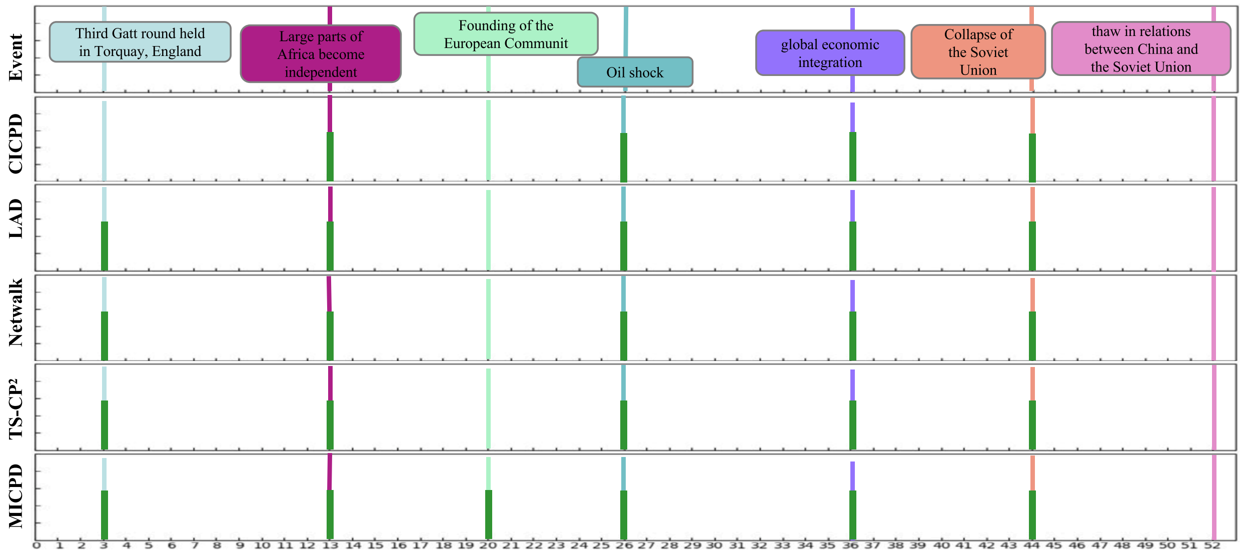


Fig. 8. Change point detection on World trade network. Detected change points are marked with green vertical bars. The events are marked in colored boxes associated with the colored lines spanning all methods.

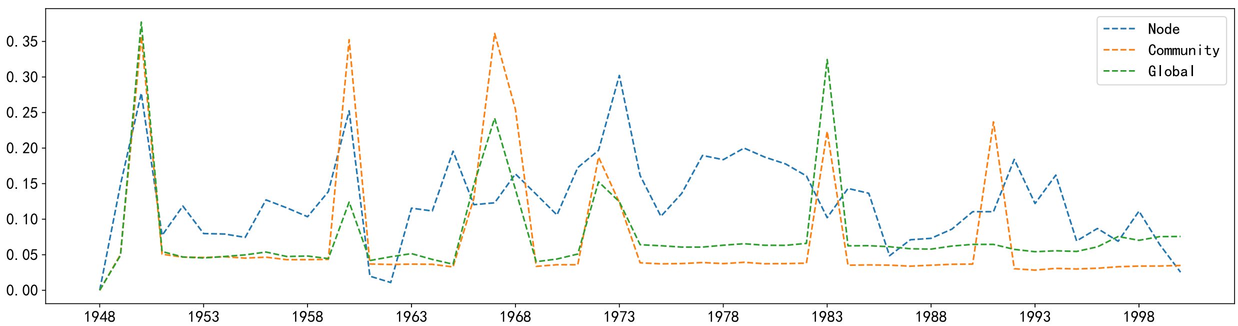
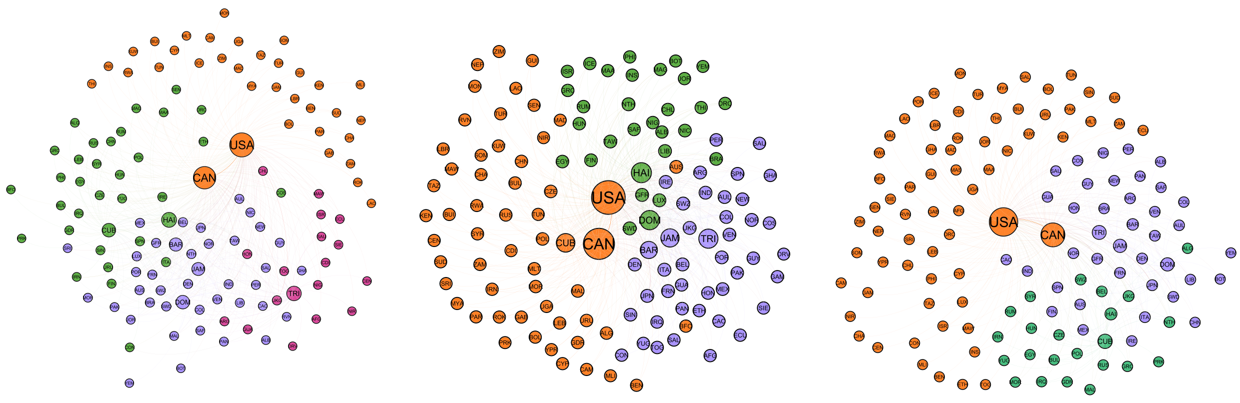
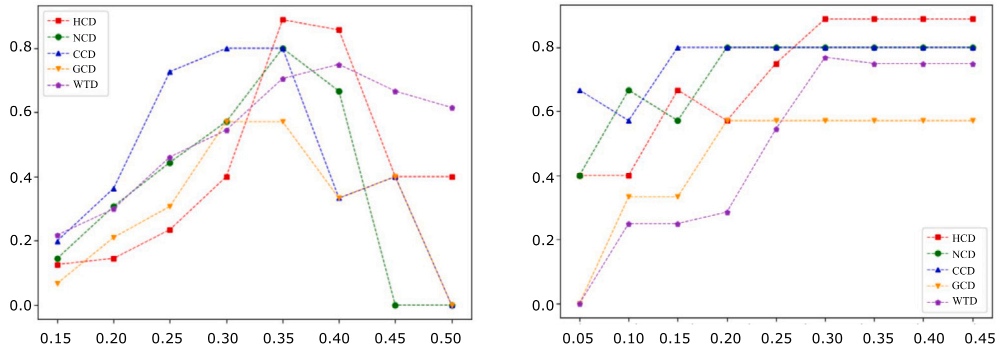


Fig. 9. Change scores from different perspectives of the World trade dataset. The blue dashed line represents the node change score, the yellow dashed line represents the community change score, and the green dashed line represents the global change score. The abscissa represents time, and the ordinate represents abnormal values. Here we normalize the abnormal values. All y axes have ranges between 0 and 1.



(a) 1966 (b) 1967 (c) 1968

Fig. 10. An change point case in World trade dataset from 1966 to 1968. Nodes represent countries, and edges represent trade between countries.



(a) th_1 (b) th_2

Fig. 11. Results of Parameters study.

performance of the synthetic dataset is not degraded because th_2 is too large, mainly because the network snapshots at non-changing time points are highly dependent on the previous.

6. Conclusion

Change point detection is a complex task because various types of changes are feasible. This paper proposes a multi-view feature interpretable change point detection framework MICPD, which locates temporal changes in large-scale evolving networks by tracking the historical evolution behavior of different objects in dynamic networks. Subsequently, all our experimental results on the five datasets show that MICPD has not only qualitative but also significant interpretative advantages. In the future, we will solve the change point detection problem in more complex dynamic attributed network with structural information and attribute information.

CRediT authorship contribution statement

Yingjie Xie: Conceptualization of this study, Methodology, Software. **Wenjun Wang:** Investigation, Resources. **Minglai Shao:** Data curation, Writing Original draft preparation. **Tianpeng Li:** Data curation, Writing - Review Editing. **Yandong Yu:** Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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